

Smart Metro Ticket Booking System

Requirement Analysis Phase

Team ID	
Project Name	Smart Metro Booking System
Maximum Marks	4 Marks

4. Technology Stack

This document outlines the technology stack proposed for the ServiceNow-based Metro Ticket Booking system, spanning presentation, application logic, integration, data, notification, and security layers.

Technology Stack — Layered Architecture



Figure 4.1 — Layered technology architecture of the ticketing solution

4.1 Layer-wise Technology Breakdown

Layer	Technology / Component	Purpose
Presentation	ServiceNow Service Portal, UI Pages, WhatsApp Business Chatbot	Passenger-facing booking form, fare preview, ticket display, and chat-based booking.
Application / Process	Service Catalog, Flow Designer, Business Rules, Script Includes, Catalog Client Scripts	Captures booking input, orchestrates the booking-to-ticket workflow, executes fare logic.

Layer	Technology / Component	Purpose
Integration	Integration Hub, REST/SOAP APIs, Payment Gateway (UPI/Card/Wallet), QR Code Generation Library	Connects the platform to external payment and QR services with logged, traceable calls.
Data	ServiceNow Custom Tables — Station Master, Fare Rule Table, Ticket/QR Records, Transaction Logs	Stores reference and transactional data required for booking, validation, and reporting.
Notification	Email Notifications, SMS Gateway, Push Notifications, WhatsApp Business API	Delivers QR tickets, payment confirmations, and exception alerts to passengers and staff.
Reporting & Analytics	Performance Analytics, Reports, Dashboards	Surfaces booking volume, revenue split, and peak-time insights for Metro Operations.
Security & Administration	Access Control Lists (ACLs), OAuth 2.0, Data Encryption, Role-based Views	Restricts data and functionality by persona and secures payment/QR data end-to-end.

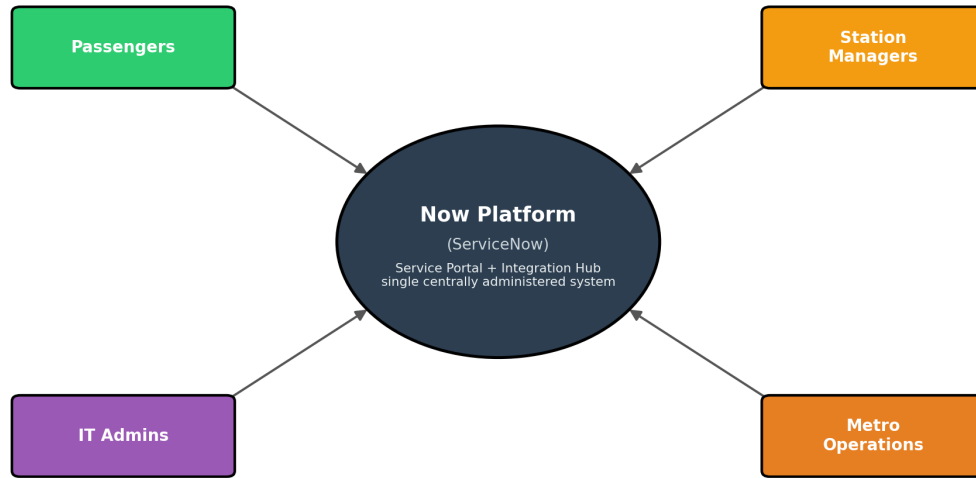
4.2 Platform Rationale

- Service Catalog provides a governed, reusable form framework for capturing structured booking input (Source, Destination, Passenger Type, Number of Tickets) with built-in client-side validation.
- Flow Designer enables a low-code, visual orchestration of the multi-step booking-to-ticket process, making the workflow easy to audit and modify without heavy scripting.
- Integration Hub standardizes outbound calls to the payment gateway and QR generation service, giving IT Admins centralized logging, retry handling, and credential management.
- Native Notifications keep ticket delivery within the ServiceNow platform, avoiding a separate messaging system while still supporting Email, SMS, Push, and WhatsApp channels.
- Performance Analytics and Reports reuse existing platform capabilities to deliver the data-driven insights Metro Operations needs for route and peak-time planning, without a separate BI tool.

4.3 Environment Considerations

The solution is designed to run entirely on the ServiceNow platform (Now Platform), using out-of-the-box Service Portal capabilities for the front end and Integration Hub spokes for all third-party connections. This keeps the technology footprint minimal, reduces licensing overhead for a separate mobile app in the initial phase, and allows all personas — Passengers, Station Managers, Metro Operations, and IT Admins — to be served from a single, centrally administered system.

Single-Platform Deployment Model



No separate mobile app or external BI/messaging system required in this phase

Figure 4.2 — Single-platform deployment model serving all four personas